DATE: 05/06/2026

ASSIGNMENT #04

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Subject :- Analysis.

PROBLEM #01 :-

Item	Weight	Value
1	3	25
2	2	20
3	1	15
4	4	40
5	5	50

$W = 6$.

Solution :-

	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	0	0	25	25	25	25
2	0	0	20	25	25	45	45
3	0	15	20	35	40	45	60
4	0	15	20	35	40	55	60
5	0	15	20	35	40	55	65

For $i = 1$ $(w_1 = 3, p_1 = 25)$;
$c = 1, 2$: Because $w_1 > c$ $(3>1, 3>2)$,
we apply Rule 3:

DATE: / /

$K(1,1) = K(0,1) = 0$
$K(1,2) = K(0,2) = 0$

- $c = 3$ to 6: Because $w_1 \le c$, we apply Rule 1;

$K(1,3) = \max[25 + K(0, 3-3), K(0,3)]$
$= \max[25+0, 0] = 25$
$K(1,4) = \max[25+0,0] = 25$
$K(1,5) = \max[25+0,0] = 25$
$K(1,6) = \max[25+0,0] = 25$

For $i=2$ $(w_2 = 2, p_2 = 20)$;
- $c = 1$: Because $w_2 > c$ $(2>1)$, apply Rule 3:
$K(2,1) = K(1,1) = 0$.

- $c = 2$ to 6: Because $w_2 \le c$, apply Rule 1:

$K(2,2) = \max[20 + K(1, 2-2), K(1,2)] = \max[20+0, 0]$
$= 20$
$K(2,3) = \max[20+0, 25] = 25$.
$K(2,4) = \max[20+0, 25] = 25$.
$K(2,5) = \max[20 + K(1, 5-2), (K(1,5)] = \max(20, 25, 25)$
$= 45$
$K(2,6) = \max[20+25, 25] = 45$.

For $i=3$ $(w_3 = 1, p_3 = 15)$:
- $c = 1$ to 6: Because $w_3 \le c$ for all remaining capacities, apply Rule 1:

DATE: / /

$K(3,1) = \max[15 + K(2,1-1), K(2,1)]$
$= \max[15 + 0, 0] = 15$

$K(3,2) = \max[15 + [K(2,2-1), K(2,2)]$
$= \max[15 + 0, 20] = 20$

$K(3,3) = \max[15 + K(2,3-1), K(2,3)]$
$= \max[15 + 20, 25] = 35$

$K(3,4) = \max[15 + 25, 25] = 40$

$K(3,5) = \max[15 + 25, 45] = 45$

$K(3,6) = \max[15 + K(2,6-1), K(2,6)]$
$= \max[15 + 45, 45] = 60$

For $i = 4$ ($w_4 = 4$, $p_4 = 40$):

- $c = 1$ to 3: Because $w_4 > c$, apply Rule 3:
  $K(4,c) = K(3,c) \Rightarrow$ Values are 15, 20, 35

- $c = 4$ to 6: Because $w_4 \leq c$ apply Rule 1:
  $K(4,4) = \max[40 + K(3,4-4), K(3,4)] = \max[40+0, 40]$
  $= 40$

  $K(4,5) = \max[40 + K(3,5-4), K(3,5)] = \max[40+15, 45] = 55$

  $K(4,6) = \max[40 + K(3,6-4), K(3,6)] = \max[40+20, 60] = 60$

For $i = 5$ ($w_5 = 5$, $p_5 = 50$):

- $c = 1$ to 4: Because $w_5 > c$, apply Rule 3:
  $K(5,c) = K(4,c) \Rightarrow$ Values are 15, 20, 35, 40

- $c = 5$ to 6: Because $w_5 \leq c$, apply Rule 1:
  $K(5,5) = \max[50 + K(4,5-5), K(4,5)] = \max[50+0, 55] = 55$

  $K(5,6) = \max[50 + K(4,6-5), K(4,6)] = \max[50+15, 60]$
  $= \max[65, 60] = 65$